

RAG Evaluation Benchmark Dataset

This document outlines the structured narrative evaluation categories for the RAG pipeline benchmark items.

Benchmark Item 1

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 2

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 3

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 4

Category	Description / Required Narrative Evaluation

User Question Category	Description / Required Narrative Evaluation
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 5

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 6

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 7

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are

Narrative Category	generated and mapped to close the lexical gap. Description / Required Narrative Evaluation
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Benchmark Item 8

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 9

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.

Benchmark Item 10

Category	Description / Required Narrative Evaluation
User Question	The query submitted by the end user to the RAG pipeline.
Target Section	The gold-standard documentation chunk or knowledge base paragraph where the true answer resides.
Expected Baseline Failure Analysis	A breakdown explaining why the standard, unoptimized baseline RAG system fails to correctly locate or answer the user question.
Expected Keyword Expansion Behavior	The designated behavior for expanding terms or token keywords to optimize semantic mapping.
Expected HyDE Retrieval Narrative	The descriptive evaluation framework summarizing how Hypothetical Document Embeddings (HyDE) are generated and mapped to close the lexical gap.